ON THE ONTIC INCOMPLETENESS OF DISCRETE DETERMINISTIC SYSTEMS

The Initial Conditions Problem, Aggregation Hierarchy, and the Ontological Separation of Law and Boundary Data, with an Application to the Past Hypothesis

C. G. Paci
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Abstract. We construct a minimal ontic logic for discrete deterministic systems and prove that any such system is ontically incomplete: its dynamics cannot determine its own initial conditions. We introduce the Void Hypothesis as the null state and show that the existence of nontrivial structure constitutes a definitive rejection of this null. We formalize an aggregation hierarchy distinguishing informational, locational, configurational, and emergent ontological levels, and prove that classical observables arise only at the aggregate level through temporal accumulation. We establish that deterministic chaos is ontically sensitive but epistemically unpredictable. From these results follows an ontological separation theorem: the complete specification of a discrete deterministic system is the ordered pair (f, C_0) of dynamical law and boundary datum, and these two structures are provably independent. Using algorithmic information theory, we prove a boundary information dominance result: for a typical initial configuration, $K(C_0) \gg K(f)$. We then prove a structural unavoidability theorem: no local translation-invariant rule can determine a generic initial configuration, because the information capacity of any such rule is bounded by a constant independent of system size. This yields a trilemma: any principle that determines C_0 must be nonlocal, translation-variant, or carry system-scale information, in every case ceasing to function as a law in the standard sense. We apply this framework to the Past Hypothesis, showing that the low-entropy initial condition is the compressible case of a general information-theoretic result. We argue that the epistemological status of these results is that of proved theorems, not falsifiable hypotheses, and that the demand for falsifiability of logical results constitutes a category error. All claims are conditional on the axioms stated (finitude, discreteness, determinism, locality, causal closure) and do not extend to systems violating these assumptions.

1. Symbolic Vocabulary

We define the minimal set of symbols required for ontic reasoning about discrete deterministic systems. No symbol carries metaphysical connotation. Each is operationally defined.

1.1 Structural Symbols

Symbol	Definition
Λ	The substrate: a finite three-dimensional cubic lattice, $\Lambda \subset \mathbb{Z}^3$
v	A vertex (site) of Λ, indexed by integer coordinates (x, y, z)
S	The state space at each vertex: $S = \{-1, 0, +1\}$
$J(v)$	The flux field at vertex v: a vector in $\mathbb{R}^3$ encoding energy density
$C(t)$	The total configuration at tick t: $C(t) = \{(s_v, J_v) \mid v \in \Lambda\}$
Ω	The configuration space: the set of all possible configurations C
f	The update map: $f : \Omega \to \Omega$, deterministic, time-independent
t	Discrete time index: $t \in \mathbb{N} = \{0, 1, 2, \ldots\}$
C_0	The initial configuration: $C(0) = C_0 \in \Omega$

1.2 Ontic Symbols

Symbol	Definition
V	The Void: the unique configuration where $s_v = 0$ and $J_v = \mathbf{0}$ for all $v \in \Lambda$
K_B	The manifestation threshold: the minimum $\|J\|$ for state transition from $s = 0$
δ	The differentiation operator: $\delta(C) = d(C, V)$, the distance from Void
$A(R,t)$	An aggregate: a connected region $R \subset \Lambda$ where $\delta > K_B$ persisting over interval $[t_0, t]$
Σ	The boundary data functional: $\Sigma := C_0$. Notational device representing the independent structure required to complete system specification.
E	External structure: any structure not contained in (Λ, f). Structure independent of the dynamical law.

1.3 Logical Connectives

Symbol	Definition
$\vdash$	Provable from within the system (internal derivability)
$\nvdash$	Not provable from within the system
$\exists!$	There exists exactly one
$\forall$	For all
$\Rightarrow$	Logically entails
$\perp$	Contradiction

2. Axioms of a Discrete Deterministic System

We state five axioms. These are not assumed for convenience. They are the minimal conditions under which a discrete deterministic system is well-defined.

> **A1 (Finitude).** *The substrate Λ is a finite subset of $\mathbb{Z}^3$. The state space S is finite. The configuration space Ω is therefore finite.*

> **A2 (Discreteness).** *Time is indexed by natural numbers. There exists no intermediate state between t and $t+1$.*

> **A3 (Determinism).** *The update map f is a function (single-valued). For every $C \in \Omega$, $f(C)$ is unique. Identical configurations produce identical successors.*

> **A4 (Locality).** *The update of each vertex v depends only on the states and fluxes of vertices within a bounded neighborhood $N(v)$.*

> **A5 (Completeness of f).** *The update map f together with C_0 fully determines all future configurations: $C(t) = f^t(C_0)$ for all $t > 0$. No additional input is required after $t = 0$.*

> **Note:** A5 is the closure axiom. It states that the system is causally closed for all $t > 0$. It says nothing about $t = 0$ itself.

Scope. All results in this paper are conditional on A1 through A5. They are theorems about discrete deterministic systems satisfying these axioms, not universal claims about all possible physical systems. If the universe is continuous, nonlocal, or nondeterministic, the theorems do not apply. The ontological claims that follow are conditional ontological claims: they describe structural properties of systems satisfying the stated axioms.

3. The Void Hypothesis

3.1 Definition

Void Hypothesis (H_0): The initial configuration is the Void: $C_0 = V$, where V is the unique configuration satisfying $s_v = 0$ and $J(v) = \mathbf{0}$ for all $v \in \Lambda$.

The Void V is the zero element of the configuration space. It represents the total absence of differentiation: no flux, no manifestation, no structure. It is the simplest possible initial condition and therefore the appropriate null hypothesis.

3.2 Properties of the Void

> **Proposition 3.1 (Void Fixed Point).** *If $f(V) = V$, then the Void is a fixed point of the dynamics. A system initialized at V remains at V for all time: $C(t) = V$ for all t.*

Proof. By A3, f is deterministic. By A5, $C(t) = f^t(C_0)$. If $C_0 = V$ and $f(V) = V$, then $C(1) = f(V) = V$. By induction, $C(t) = V$ for all $t \geq 0$. □

A void lattice with no flux generates no flux. Emptiness propagates emptiness. This is the ontic analog of Newton's first law applied to the substrate itself.

3.3 Rejection of the Void Hypothesis

> **Proposition 3.2 (Rejection).** *The observed universe is not the Void. Therefore H_0 is false. Therefore $C_0 \neq V$.*

Proof. By empirical observation, there exist configurations with $s_v \neq 0$ and $|J(v)| > 0$. That is, $\delta(C(t_{\text{now}})) > 0$ for the present configuration. If H_0 were true and $f(V) = V$, then $C(t) = V$ for all t, contradicting observation. Therefore H_0 is rejected. □

This is the most elementary empirical result available: something exists. The mere existence of nonzero structure at any time $t > 0$, given Proposition 3.1, is sufficient to conclude that $C_0 \neq V$. We have rejected the null.

4. Ontic Incompleteness

4.1 Statement

Theorem 4.1 (Ontic Incompleteness). *Let (Λ, S, f, Ω) be a discrete deterministic system satisfying A1 through A5. Then the initial configuration C_0 is not derivable from f. Formally: $f \nvdash C_0$.*

Proof. Suppose for contradiction that $f \vdash C_0$. That is, suppose C_0 can be derived from the update map f alone.

By A3, f maps configurations to configurations: $f : \Omega \to \Omega$. For f to determine C_0, there must exist some configuration C^* such that $f(C^*) = C_0$, and C^* must itself be determined by f.

But C^* is a configuration at time $t = -1$. By A2, time is indexed by natural numbers: $t \in \mathbb{N}$. There is no $t = -1$. The domain of the dynamical history is $\{C(0), C(1), C(2), ...\}$. There is no predecessor to C_0 within the system.

Therefore the assumption that $f \vdash C_0$ requires the existence of a state at a time that does not exist. This is a contradiction. Therefore $f \nvdash C_0$. □

4.2 Consequences

Corollary 4.2 (Irreducible External Dependence). The specification of any discrete deterministic system requires exactly one element not determined by the system: the initial configuration C_0.

Corollary 4.3 (Contrast with Continuum Systems). In continuous-time systems, the initial conditions problem is often obscured by gauge freedom, boundary conditions at infinity, or probabilistic reinterpretation. In a discrete deterministic system, the problem is sharp: there is a first tick, it has a definite configuration, and nothing within the dynamics produces it.

4.3 Relation to Gödel

Theorem 4.1 is structurally analogous to Gödel's first incompleteness theorem, but operates in a different domain. Gödel showed that a sufficiently powerful formal system cannot prove all truths about itself. Theorem 4.1 shows that a deterministic dynamical system cannot produce its own initial state. The parallel is precise: in both cases, a self-contained system requires input from outside itself to be fully specified.

The difference is that Gödel's result is about provability within formal logic. Ours is about causation within dynamics. We call this *ontic incompleteness* to distinguish it from logical incompleteness.

5. The Aggregation Hierarchy

We formalize the ontological levels that arise between the raw substrate and observable physics.

5.1 Four Ontological Levels

Level 0: Informational. A vertex $v \in \Lambda$ considered purely as an address. It has coordinates $(x, y, z) \in \mathbb{Z}^3$ but no state assignment. At this level, v is epistemic: it encodes where something could be, not that something is.

Level 1: Locational. A vertex v equipped with a definite state assignment (s_v, J_v). The vertex now has physical content. But a single located vertex has no emergent properties. It has no wave behavior, no temperature, no mass in the macroscopic sense. It is a datum, not a phenomenon.

Level 2: Configurational (Aggregate). A connected region $R \subset \Lambda$ containing N vertices, where the collective flux exceeds K_B for the aggregate. At this level, the collection persists across multiple ticks. This is the first level at which identity is meaningful: the aggregate can be tracked, measured, and distinguished from the background. We call this a *particle*.

Level 3: Emergent. Properties arising from the temporal evolution of aggregates that have no analog at Levels 0 through 2. These include wave behavior, thermodynamic quantities, classical trajectories, and measurement outcomes. These properties require both spatial aggregation and temporal accumulation.

5.2 The Emergence Theorem

> **Theorem 5.1 (No Premature Emergence).** *Let P be any Level 3 property (wave function, temperature, classical trajectory). Then P is not definable at Level 1. P requires both spatial extent ($|R| \gg 1$) and temporal depth ($t \gg 1$).*

Proof. Level 3 properties are defined as functions of coarse-grained fields: $P = F(\bar{J}(X, T))$ where $\bar{J}$ is the spatiotemporal average over a block $B(X)$ and time interval $I(T)$. For $|B| = 1$ and $|I| = 1$, no averaging occurs, and P reduces to a Level 1 datum. The coarse-grained field equations require spatial and temporal derivatives defined only when the averaging domain has extent greater than one in both dimensions. □

5.3 The Voxel as Ontological Unit

We define the voxel as a vertex at Level 1: a lattice site equipped with state and flux. The voxel is the minimal ontological unit. It is not a particle. It is not a wave. It is the substrate from which particles and waves emerge through aggregation and temporal evolution.

Definition 5.2. A *voxel* is an element $v \in \Lambda$ together with its assigned state (s_v, J_v) at a given tick t. A voxel is *active* if $|J_v| > 0$ and *manifested* if $|J_v| > K_B$.

Definition 5.3. A *particle* is an aggregate $A(R, t)$ where (i) R is a connected subregion of Λ, (ii) the total activation exceeds K_B, (iii) the aggregate persists over a time interval, and (iv) it is distinguishable from the background.

Particles do not exist at Level 0 or Level 1. They are Level 2 entities. This resolves the conceptual error in point-particle physics, which attempts to assign Level 2 and Level 3 properties (mass, charge, spin, wave function) to Level 0 objects (dimensionless points).

6. Determinism, Stochasticity, and the Dual Nature of Chaos

6.1 Deterministic Stochasticity

> **Proposition 6.1 (Deterministic Uniqueness).** *For a system satisfying A1 through A5, the trajectory $\{C(0), C(1), C(2), \ldots\}$ is unique given C_0. There is exactly one history for each initial condition.*

Proof. Immediate from A3 and A5 by induction on t. □

The trajectory is not random. It is not probabilistic. However, from the perspective of any observer embedded within the system, the trajectory may appear stochastic for the following reason.

> **Proposition 6.2 (Epistemic Stochasticity).** *An observer embedded in (Λ, f, Ω) cannot determine C_0 with infinite precision, because the observer is itself a finite aggregate within Λ. Therefore the observer cannot predict the full trajectory, even though the trajectory is fully determined.*

Proof. The observer is a subsystem $O \subset \Lambda$. To determine C_0 completely, the observer would need to encode every state and flux value of every vertex in Λ, including the vertices comprising the observer itself. Since O is a proper subset of Λ, the observer's information capacity is strictly less than the system's information content. □

6.2 Chaos: Ontic Sensitivity, Epistemic Unpredictability

Definition 6.3. *Deterministic chaos* is the property that nearby initial conditions produce exponentially diverging trajectories. Formally: for C_0 and C_0' with $d(C_0, C_0') = \varepsilon$, there exists $\lambda > 0$ such that $d(C(t), C'(t))$ grows as $e^{\lambda t}$.

> **Proposition 6.4 (Dual Nature of Chaos).** *Chaos has both ontic and epistemic components. The dynamical sensitivity (positive Lyapunov exponent, mixing, ergodicity) is a property of the map f and is ontic: it exists independent of any observer. The unpredictability is epistemic: it arises from the observer's inability to specify C_0 with sufficient precision. These are distinct properties and must not be conflated.*

Proof. The Lyapunov exponent $\lambda = \lim_{t\to\infty} (1/t) \ln |\mathrm{d}f^t/\mathrm{d}C|$ is a property of f alone, computable without reference to any observer. Therefore sensitivity is ontic. Predictability requires an agent with a model and finite-precision initial data. By Proposition 6.2, any embedded observer has information capacity strictly less than the system's information content. When $\lambda > 0$, initial measurement error ε grows as $\varepsilon e^{\lambda t}$, exceeding any fixed tolerance in finite time. Therefore unpredictability is epistemic. □

The common phrase "chaos is random" is a category error. It attributes an epistemic property (apparent randomness) to the system's ontology (which is fully determined). But the corrected statement is not that chaos is "merely subjective." The Lyapunov exponent is as objective as the gravitational constant. What is subjective is the inference from sensitivity to randomness.

7. The Ontological Separation Theorem

7.1 The Argument

We combine the results of Sections 3 through 6 into a single logical chain.

Premise 1 (Ontic Incompleteness). $f \nvdash C_0$. The dynamics do not determine the initial configuration (Theorem 4.1).

Premise 2 (Void Rejection). $C_0 \neq V$. The null hypothesis is empirically rejected (Proposition 3.2).

Premise 3 (No Internal Randomness). The system is deterministic (A3). There is no internal random number generator. No probability distribution over Ω is derivable from f.

> **Theorem 7.1 (Ontological Separation).** *The complete specification of a discrete deterministic system is the ordered pair (f, C_0), where C_0 is logically independent of f. No reduction of this pair to a single structure is possible within the system.*

Proof. By Premise 1, $f \nvdash C_0$: the dynamics do not entail the initial configuration. By Premise 2, $C_0 \neq V$: the initial configuration is nontrivial. By Premise 3, no probability measure over Ω is generated by f, so C_0 is not a dynamically-generated random variable. Therefore the specification of the system requires two independent inputs: (i) the dynamical law f, which determines all transitions $C(t) \to C(t{+}1)$, and (ii) the boundary datum C_0, which determines the starting point. Neither is derivable from the other. □

7.2 Structural Analogies

Theorem 7.1 is not novel in its mathematical content. It is the discrete, ontological formulation of a separation that appears throughout mathematical physics. A PDE determines the evolution of a field but not its boundary values. The Hamiltonian determines the flow on phase space but not the initial point. The Schrödinger equation determines the evolution of the wave function but not the wave function at $t = 0$. What Theorem 7.1 adds is the ontological interpretation: for a system that is fundamentally discrete and deterministic, the separation between law and boundary data is a structural property of reality itself.

7.3 The Notation Σ

Definition 7.2. We write $\Sigma := C_0$ as a notational device to denote the boundary datum. Σ is not a function, operator, or entity. It is a label for the second member of the ordered pair (f, C_0). We introduce it solely to have a symbol for the structure that f does not provide.

This is deliberate understatement. The question of what determines C_0 is not addressed by the theorem and is not representable within the formalism. The theorem establishes that C_0 is independent of f. It does not, and cannot, explain why C_0 takes the value it does.

7.4 Boundary Information Content

We now quantify the informational content of C_0 using algorithmic information theory.

Definition 7.3 (Kolmogorov Complexity). The Kolmogorov complexity $K(x)$ of a finite object x is the length of the shortest program that produces x on a universal Turing machine. $K(x)$ measures the incompressibility of x: the minimum number of bits required to specify x.

> **Proposition 7.4 (Typical Boundary Information).** *For a finite configuration space Ω with $|\Omega|$ elements, the fraction of configurations C with $K(C) < \log_2|\Omega| - k$ is at most 2^{-k}. That is, for a typical configuration, $K(C_0) \approx \log_2|\Omega|$.*

Proof. Standard result in algorithmic information theory (Kolmogorov 1965, Chaitin

1966). The number of programs of length less than m is at most $2^m - 1$. Therefore the number of objects with $K(x) < m$ is at most $2^m - 1$. Setting $m = \log_2|\Omega| - k$, the fraction of Ω with complexity below this threshold is at most 2^{-k}. □

Remark on representation dependence. Kolmogorov complexity $K(x)$ depends on the choice of universal Turing machine U, but only up to an additive constant c_U that is independent of x and independent of $|\Lambda|$. Since the dominance result $K(C_0) \gg K(f)$ concerns the asymptotic behavior as $|\Lambda| \to \infty$, and $K(f)$ is bounded by a constant while $K(C_0)$ grows linearly with $|\Lambda|$, the additive constant does not affect the conclusion. All complexity comparisons in this paper hold up to $O(1)$ terms that vanish in the relevant limit.

Theorem 7.5 (Boundary Information Dominance). *For a typical initial configuration, $K(C_0) \approx \log_2|\Omega|$, while $K(f) = O(|N|)$ where $|N|$ is the neighborhood size in A4. Therefore $K(C_0) \gg K(f)$ for any nontrivial lattice. In systems satisfying A1 through A5, the majority of informational degrees of freedom reside in boundary data rather than in dynamical law.*

Proof. The update map f is specified by a local, translation-invariant lookup table of size $|S|^{|N|} \times |S|$, which is a constant independent of lattice size. Therefore $K(f) = O(3^{|N|} \log 3)$. The configuration space $|\Omega| = 3^{|\Lambda|}$, so $\log_2|\Omega| = |\Lambda| \log_2 3$, which grows linearly with lattice size. For any lattice with $|\Lambda| \gg 3^{|N|}$, the boundary datum carries overwhelmingly more information than the dynamical law. □

This is the quantitative content of the ontological separation. The dynamical law f is simple: a finite, local, translation-invariant rule. The boundary datum C_0 is (generically) complex: an incompressible specification of the state of every vertex. In discrete deterministic systems satisfying A1 through A5, the majority of informational degrees of freedom reside in boundary data rather than in dynamical law.

7.5 The Compressible Case

A referee may object: perhaps our C_0 is not typical. Perhaps $K(C_0)$ is much less than $\log_2|\Omega|$. This is logically possible and would mean the boundary datum carries less information than the generic case.

Note that this objection does not weaken Theorem 7.1 (Ontological Separation), which holds regardless of the complexity of C_0. What it would affect is Theorem 7.5: if C_0 is highly compressible, the boundary datum is still independent of f but carries less total information.

However, compressibility of C_0 creates its own puzzle. If $K(C_0)$ is much less than $\log_2|\Omega|$, then C_0 is a member of a vanishingly small subset of Ω. A note of precision: low thermodynamic entropy (small macrostate volume) and low Kolmogorov complexity (short description) are distinct concepts. A microstate within a small macrostate may still be algorithmically random. The claim here concerns the description length of the macrostate constraint, which bounds $K(C_0)$ from above, not the internal randomness of C_0 within its macrostate. With this distinction, compressibility is precisely the low-entropy condition identified by Penrose. A compressible C_0 is a structured C_0, and the question of why C_0 has this particular structure replaces the question of why C_0 has this particular value. Either way, the boundary datum demands explanation that f cannot provide.

Summary of cases: (a) If $K(C_0) \approx \log_2|\Omega|$ (generic), the boundary datum dominates the informational content of the system. (b) If $K(C_0)$ is much less than $\log_2|\Omega|$ (compressible), the boundary datum is atypical, implying additional constraints beyond f and beyond uniform sampling. In either case, C_0 is independent of f and constitutes non-dynamical structure requiring specification.

7.6 Structural Unavoidability

A natural objection to Theorem 7.5 is: perhaps a deeper law g, not yet identified, determines both f and C_0. If so, the ordered pair (f, C_0) would reduce to a single structure, and boundary information dominance would be an artifact of our ignorance. We now show that this escape is formally constrained.

Lemma 7.6 (Information Capacity of Local Laws). Let g be any rule on a lattice $\Lambda \subset \mathbb{Z}^3$ with state space S and neighborhood radius r that is (i) local and (ii) translation-invariant. Then $K(g) \leq C(|S|, r)$, where C is a constant depending on $|S|$ and r but independent of $|\Lambda|$.

Proof. A local translation-invariant rule is fully specified by a lookup table mapping neighborhood configurations to outputs. The neighborhood of radius r in $\mathbb{Z}^3$ contains $|N(r)| = O(r^3)$ sites. The table has $|S|^{|N(r)|}$ entries, each requiring $\log_2|S|$ bits. For $S = \{-1, 0, +1\}$ and $r = 1$, the neighborhood contains at most 27 sites. The lookup table has $3^{27} \approx 7.6 \times 10^{12}$ entries: a large but fixed number independent of $|\Lambda|$. Therefore $K(g) = O(1)$ with respect to lattice size. □

> **Theorem 7.7 (Structural Unavoidability).** *No local translation-invariant rule can determine the initial configuration of a nontrivial lattice. If g is local and translation-invariant, then $K(g) = O(1)$, but for a typical C_0, $K(C_0) \sim |\Lambda| \log_2|S|$. Since $O(1) \ll O(|\Lambda|)$ for any $|\Lambda| \gg 1$, no such g can encode C_0. Boundary information dominance is structurally unavoidable.*

Proof. Direct from Lemma 7.6 and Proposition 7.4. $K(g)$ is bounded by a constant independent of $|\Lambda|$. $K(C_0)$ grows linearly with $|\Lambda|$. No amount of ingenuity in the design of g can overcome the linear gap between a constant and a quantity that scales with system size. □

> **Corollary 7.8 (The Trilemma).** *Any principle that fully determines C_0 for a nontrivial lattice must satisfy at least one of: (a) it is nonlocal, (b) it breaks translation invariance, or (c) its description length scales with $|\Lambda|$. In cases (a) and (b), the principle violates the defining properties of a physical law as understood in local field theory. In case (c), the principle is functionally equivalent to boundary data.*

Proof. By Lemma 7.6, any principle satisfying both locality and translation invariance has $K = O(1)$. To encode $K(C_0) \sim |\Lambda|$ bits, one of these constraints must be violated or the description length must scale. These three options are exhaustive. □

7.7 Logical Closure of the Trilemma

We trace each branch to its terminal conclusion.

Branch (a): *g* is nonlocal. The output at site v depends on the states of arbitrarily distant sites. To determine C_0 at v, the principle g must access $O(|\Lambda|)$ bits of global configuration data. It is boundary data in a distributed representation, not a law in the standard sense.

Sub-case: bounded nonlocality. A referee may propose that g has range R > 1 but R is fixed, independent of $|\Lambda|$. This is "nonlocal" in common parlance but local in the sense of Lemma 7.6: the neighborhood contains $O(R^3)$ sites, and $K(g) = O(|S|^{O(R^3)})$, which is a constant independent of lattice size. Lemma 7.6 already covers this case. If R is fixed, boundary dominance holds. If R scales with $|\Lambda|$, then $K(g)$ scales with $|\Lambda|$, and we are in Branch (c). There is no intermediate case.

Branch (b): *g* breaks translation invariance. Different rules apply at different sites. Then g requires a site-specific specification: either $|\Lambda|$ distinct lookup tables or one table plus $|\Lambda|$ site labels. In either case, $K(g) = O(|\Lambda|)$. The site-specific variation is itself the boundary data. A critic who asserts that the universe is not translation-invariant at the fundamental level is not escaping the conclusion but restating it: the translation-variant structure is precisely the system-scale information that the theorem identifies as boundary data.

Branch (c): *K*(*g*) scales with |Λ|. The principle carries information proportional to system size. Its information content is that of boundary specification. Relabeling it does not change $K(g)$.

The trichotomy is exhaustive because Lemma 7.6 identifies exactly two properties that make $K(g) = O(1)$: locality and translation invariance. Violating either one, or allowing the description length to grow, covers all remaining possibilities. There is no fourth branch.

7.8 The Incompleteness of Law-Primary Ontologies

> **Corollary 7.9 (Informational Incompleteness).** *In any discrete deterministic system satisfying A1 through A5 with local translation-invariant dynamics, the dynamical law f accounts for at most $O(1)/O(|\Lambda|) = O(1/|\Lambda|)$ of the total informational content. Any ontological framework that treats laws as the primary explanatory structure is informationally incomplete: it addresses a fraction of the system's specification that vanishes as the system grows.*

Proof. The total informational content is $K(f) + K(C_0)$ (up to $O(1)$ terms). $K(f) = O(1)$ by Lemma 7.6. $K(C_0) = O(|\Lambda|)$ for a typical C_0. The fraction attributable to f is $O(1/|\Lambda|) \to 0$ as $|\Lambda| \to \infty$. □

This is not a philosophical claim. It is an inequality. The informational contribution of the law is bounded by a constant. The informational contribution of the boundary datum grows without bound. For any physical lattice of cosmological size, the law accounts for a negligible fraction of the system's specification. Ontologies that privilege laws over boundary data have the information content inverted.

7.9 Intellectual Lineage

The separation between dynamics and initial conditions has been recognized, in various forms, by:

Von Neumann (1932): Distinguished the unitary evolution of the quantum state from the initial preparation, identifying these as logically independent operations on the Hilbert space.

Gödel (1931): Showed that sufficiently powerful formal systems cannot derive all truths about themselves. Theorem 4.1 is the dynamical analog: a deterministic system cannot derive its own boundary data.

Kolmogorov (1965), Chaitin (1966): Established algorithmic information theory, providing the tools to quantify the incompressibility of boundary data.

Penrose (1989): Identified the low-entropy initial condition as a fundamental puzzle requiring explanation beyond the dynamical laws.

The present contribution is not new mathematics in any of its individual steps. It is the unification of structural independence (Theorem 7.1), algorithmic dominance (Theorem 7.5), and structural unavoidability (Theorem 7.7) into a single framework. The unavoidability result, in particular, appears to be new: the observation that no local translation-invariant principle can bridge the $O(1)$ versus $O(|\Lambda|)$ gap has not, to our knowledge, been stated as a formal theorem in the foundations literature.

8. The First Perturbation

8.1 Definition

Definition 8.1. The *first perturbation* is the element $P_0 = C_0 - V$: the difference between the initial configuration and the Void. P_0 encodes all nontrivial content of the boundary datum.

Since V is the zero of the configuration space, P_0 is equivalent to C_0 itself. But the notation P_0 emphasizes the structural content: P_0 is what distinguishes our universe from emptiness.

8.2 Sufficiency

> **Proposition 8.2 (Sufficiency of the Pair).** *The ordered pair (f, C_0) determines the complete history of the system. No additional input is needed.*

Proof. Immediate from A5. $C(t) = f^t(C_0)$ for all $t > 0$. □

The entire dynamical content of a discrete deterministic universe is encoded in two independent objects: a rule f and a datum C_0. All complexity, all structure, all emergence follows from the iterative application of f to C_0. The pair is both necessary and sufficient.

9. Application: The Past Hypothesis

The results of Sections 4 through 8 are purely structural. We now apply them to one specific physical question. This section is empirical and separable from the preceding logical results.

9.1 The Past Hypothesis

The Past Hypothesis (Albert 2000, following Boltzmann) states that the universe began in a state of extraordinarily low thermodynamic entropy. In our notation: the observed C_0 lies in a subset $S_{\text{low}} \subset \Omega$ with $|S_{\text{low}}|$ much smaller than $|\Omega|$.

A clarification on the relationship between entropy and complexity is required. Low thermodynamic entropy means C_0 belongs to a macrostate of small phase-space volume. This does not imply that every microstate within that macrostate has low Kolmogorov complexity; individual microstates may still be algorithmically random. However, the macrostate constraint itself has a description length of approximately $\log_2(|\Omega|/|S_{\text{low}}|)$ bits. It is this constraint, not the microstate detail, that constitutes the compressible structure.

With this distinction in hand, the Past Hypothesis maps to the compressible case of Section 7.5. This has two consequences.

9.2 Consequences

Consequence 1 (Non-genericity). C_0 is atypical with respect to the uniform measure on Ω. The set of low-entropy configurations has measure approaching zero for large $|\Lambda|$. Therefore C_0 was not produced by uniform sampling.

Consequence 2 (Structured boundary). Since $K(C_0)$ is much less than $\log_2|\Omega|$, the boundary datum has internal structure: patterns, symmetries, or regularities. This structure is not explained by f (Theorem 7.1) and is not explained by typicality (Proposition 7.4). It constitutes a third type of information: neither law nor noise, but structured boundary data.

9.3 The Boundary Information Spectrum

Combining Theorem 7.5 with the Past Hypothesis yields a spectrum:

$$K(f) \ll K(C_0) \ll \log_2|\Omega|$$

The dynamical law f is simple. A generic boundary datum would be maximally complex. Our universe's boundary datum lies between these extremes: complex enough to encode all observable structure, yet structured enough to be far from generic. Whether this intermediate complexity admits further compression is the central open question of fundamental cosmology, restated here in information-theoretic terms.

10. On the Epistemological Status of Proved Theorems

A methodological remark is required. The results of this paper fall into two distinct epistemological categories, and confusing them is a pervasive error in contemporary philosophy of science. We address it directly.

10.1 The Category Error

Since Popper (1934), the demarcation criterion of falsifiability has been widely adopted as the standard by which claims earn the status of "scientific." This adoption has been productive for empirical hypotheses. It has also bred an intellectual habit that is less productive: the reflexive demand that *all* results in a scientific paper be falsifiable, including logical and mathematical ones.

This demand is a category error. Falsifiability is a criterion for empirical hypotheses. It was never intended, and does not function, as a criterion for logical theorems. Popper himself was explicit on this point: mathematical truths are not scientific hypotheses and do not require empirical falsification. Demanding falsifiability of a theorem is structurally identical to demanding empirical evidence that 2 + 2 = 4. The question does not apply.

We state this because the results of Sections 4 through 7 are *proved*. They are not conjectures. They are not models. They are not hypotheses awaiting experimental confirmation. Theorem 4.1 (Ontic Incompleteness) follows from the axioms by contradiction. Theorem 7.1 (Ontological Separation) follows from Theorem 4.1 and the observation that something exists. Theorem 7.7 (Structural Unavoidability) follows from a counting argument on lookup tables. These results cannot be falsified because they are not empirical claims. They can be *invalidated* only by rejecting one or more of the axioms A1 through A5, which is a different operation entirely.

10.2 Proof, Falsification, and the Hierarchy of Warrant

The epistemological hierarchy is as follows, in descending order of warrant:

(i) Logical proof. A result that follows by valid inference from stated axioms. It holds with certainty conditional on those axioms. It cannot be falsified by any observation because it does not make an empirical claim. It can be invalidated only by showing the axioms are inconsistent or inapplicable. Examples in this paper: Theorems 4.1, 7.1, 7.5, 7.7, Corollaries 7.8 and 7.9.

(ii) Empirical falsification. A result that makes a definite prediction about observable quantities. It can be tested against experiment and, if the prediction fails, rejected. This is the domain of Popper's criterion, and the criterion is appropriate here. Examples in this paper: the Void Hypothesis (falsified), the aggregation hierarchy predictions, the Past Hypothesis application.

(iii) Speculative interpretation. A result that goes beyond both proof and prediction to offer an interpretation of the formal structure. It is neither provable nor falsifiable but may be productive or sterile. Example: the question of what determines C_0.

The error we identify is the conflation of (i) and (ii): the assumption that because a result appears in a physics paper, it must be an empirical hypothesis subject to falsification. This assumption is false. A physics paper may contain theorems, and those theorems have the epistemological status of proofs, not predictions. To demand falsifiability of a proof is not rigor. It is a misapplication of a criterion to a domain where it has no jurisdiction.

10.3 The Arrogance of the Criterion

We go further. The elevation of falsifiability to the sole arbiter of scientific legitimacy has had a corrosive effect on foundational inquiry. It has created an intellectual culture in which structural results are dismissed because they are "not testable," while empirical results of no structural significance are celebrated because they are. This is a distortion of the scientific enterprise.

Consider: the Pythagorean theorem is not falsifiable. No experiment can refute it. Its truth follows from the axioms of Euclidean geometry. If space is non-Euclidean, the theorem does not "fail"; it simply does not apply. No one considers this a deficiency of the theorem. The theorem remains a permanent contribution to human understanding precisely because it is proved rather than observed.

Theorem 7.7 (Structural Unavoidability) occupies an analogous position. It is not a claim about the universe. It is a claim about any discrete deterministic system with local translation-invariant dynamics. If the universe is not such a system, the theorem does not fail; it does not apply. If the universe is such a system, the theorem is not "confirmed by evidence"; it is guaranteed by the structure of the system itself.

Falsifiability identifies which claims are empirical. It does not identify which claims are true, important, or worth knowing. A proved theorem about the information-theoretic structure of deterministic systems is a permanent contribution to knowledge. An unfalsified but poorly motivated empirical hypothesis is not. The hierarchy of warrant is proof first, falsification second, speculation third. We do not accept the inversion of this hierarchy, and we make no apology for producing results that are proved rather than merely not-yet-refuted.

11. Falsifiability and Empirical Content

Having established the epistemological framework in Section 10, we now classify the specific claims of this paper.

11.1 Logical Claims (Proved)

Theorem 4.1 (Ontic Incompleteness) is a logical theorem. It cannot be falsified because it is proved from the axioms. It can be invalidated only by rejecting one of A1 through A5.

Theorem 7.1 (Ontological Separation) follows from Theorem 4.1 and the empirical rejection of H_0. It is as secure as the axioms and the observation that something exists.

Theorem 7.5 (Boundary Information Dominance) follows from standard results in algorithmic information theory applied to finite configuration spaces.

Theorem 7.7 (Structural Unavoidability) follows from a counting argument on lookup tables for local translation-invariant rules. It can be invalidated by showing that the universe's dynamics are nonlocal or translation-variant, in which case A4 would be rejected.

11.2 Empirical Claims (Falsifiable)

The Void Hypothesis H_0 is empirically falsifiable and has been falsified: something exists.

The aggregation hierarchy (Section 5) predicts that point-particle idealizations break down at sufficiently high energy. If particles are confirmed to be structureless to arbitrarily small scales, the hierarchy is falsified.

The dual nature of chaos (Proposition 6.4) predicts that dynamical sensitivity is measurable and objective, while long-term predictability remains bounded by initial-condition precision.

The cosmological application (Section 9) predicts that the information-theoretic complexity of the initial condition satisfies $K(f) \ll K(C_0) \ll \log_2|\Omega|$. If all observable structure can be derived from a dynamical law alone ($K(C_0) = 0$), the framework is falsified.

11.3 Boundary Claims (Open)

The interpretation of C_0 is not determined by the present analysis. The independence of C_0 from f is a theorem. Whether C_0 admits further explanation, and what form such an explanation might take, is an open question.

12. Summary of Results

We have established the following chain of results using only the axioms of a discrete deterministic system and the single empirical observation that the universe is not void.

(i) The Void is the unique null state and, under generic dynamics, a fixed point.

(ii) The Void Hypothesis is empirically rejected by the existence of structure.

(iii) The initial configuration is not derivable from the dynamics (Ontic Incompleteness). The system is ontically open at $t = 0$.

(iv) Classical observables arise only at the aggregate level through temporal accumulation (Emergence Theorem).

(v) Deterministic chaos is ontically sensitive but epistemically unpredictable. Determinism does not entail predictability.

(vi) The complete specification of a discrete deterministic system is the ordered pair (f, C_0), where C_0 is independent of f (Ontological Separation).

(vii) For a typical C_0, the boundary datum carries overwhelmingly more information than the dynamical law: $K(C_0) \gg K(f)$ (Boundary Information Dominance).

(viii) No local translation-invariant rule can determine a generic initial configuration (Structural Unavoidability). Any principle that does so must be nonlocal, translation-variant, or carry system-scale information (Trilemma). In every case, it is functionally equivalent to boundary data.

(ix) Any ontological framework that treats laws as the primary explanatory structure is informationally incomplete: the law accounts for a fraction $O(1/|\Lambda|)$ of the total specification, which vanishes as the system grows.

(x) If C_0 is atypical (compressible/low-entropy), the boundary datum has internal structure not explained by either f or typicality. This is the information-theoretic content of the Past Hypothesis.

(xi) The interpretation of C_0 is an open question. The separation theorem is agnostic among brute fact, law-level constraint, block universe, multiverse, and external agency.

The central thesis, stated precisely: in discrete deterministic systems satisfying A1 through A5 with local translation-invariant dynamics, boundary information dominance is structurally unavoidable. The law accounts for $O(1)$ bits. The boundary datum accounts for $O(|\Lambda|)$ bits. No deeper local law can close this gap. Any ontological framework that treats dynamical law as the primary explanatory structure is therefore informationally incomplete, and this incompleteness is not a feature of our descriptions but of the logical structure of such systems.

12.1 Limitations and Conditionality

The results are conditional on A1 through A5. Three scenarios in which the conclusions would weaken or fail should be stated explicitly.

(a) Continuity. If the universe is fundamentally continuous, the clean separation at $t = 0$ may be replaced by gauge freedom or boundary conditions at infinity, and the Kolmogorov complexity framework does not apply in the same form.

(b) Nonlocality or global law. If the dynamical law encodes global data (violating A4), the complexity $K(f)$ may scale with lattice size, and the separation may collapse.

(c) Lawful initial conditions. If C_0 is determined by a deeper principle, the ordered pair would reduce to a single structure. However, Theorem 7.7 and Corollary 7.8 formally constrain this possibility: any such deeper principle must be nonlocal, translation-variant, or carry system-scale information. In every case, it ceases to function as a law in the standard sense and becomes functionally equivalent to boundary data. The escape hatch exists in principle but leads back to the same conclusion by a different route.

Acknowledging these possibilities does not weaken the results. It localizes them: the theorems hold precisely when and where the axioms hold, and the contribution is to have made this localization explicit.

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